Week 6 Matrix Estimation

Matrix estimation: recover a matrix from linear measurements.

Problem 1: matrix completion. Suppose Θ is a low rank matrix, Y is a set of incomplete measurements. Want: $\hat{\Theta}(Y)$.

$$\Theta = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \xrightarrow{Y = A(\Theta)} Y = \begin{bmatrix} 1 & 2 \\ 2 & ? \end{bmatrix}$$

low rank of $\Theta \Rightarrow \Theta_{22} = \frac{\Theta_{12}\Theta_{21}}{\Theta_{11}} = 4$

Problem 2 (more general than problem 1): Θ low rank, Y is my set of linear measurements of Θ. Want $\hat{\Theta}(Y)$

$$Y_1 = A_1(\Theta) = \langle \Theta, A_1 \rangle = \mathrm{Tr}(\Theta^T A_1) \in \mathbb{R}$$

$$\vdots$$

$$Y_n = \mathrm{Tr}(\Theta^T A_n) \in \mathbb{R}$$

Problem 3 : Multivariate Regression.

$$Y = X\Theta^* + E$$

$\Theta^* \in \mathbb{R}^{d \times T}$: unknown matrix

$X \in \mathbb{R}^{n \times d}$: design matrix

$E \in \mathbb{R}^{n \times T}$: noise matrix

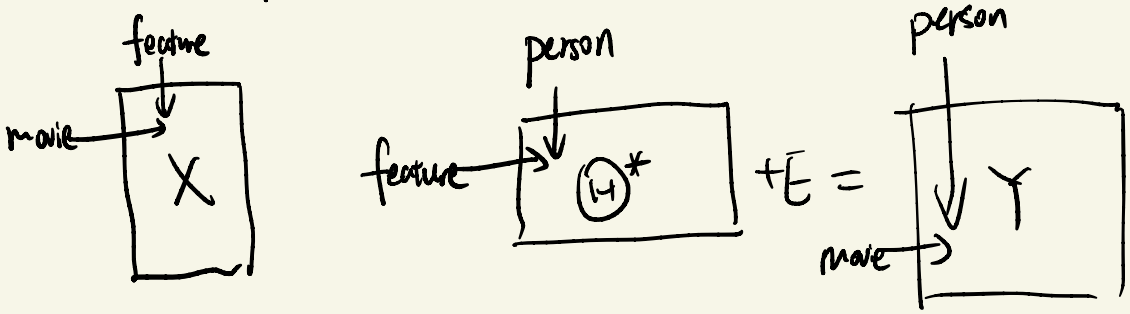
$Y \in \mathbb{R}^{n \times T}$: observation

want : $\hat{\Theta}(Y, X)$

Example : n : # movies

d : # features (duration, music quality, plot --)

T : # persons



- Note : Θ^* may contain many zero rows

  suppose k = # nonzero rows of Θ^*

  $k \geq \text{rank}(\Theta^*)$

$$\|A\|_F := \sqrt{\sum_{ij} A_{ij}^2} = \sqrt{\mathrm{Tr}(A^TA)} \quad \Rightarrow \quad \|A\|_F^2 = \sum_j \left(\sum_i A_{ij}^2\right)$$

Use Lasso estimator in Chap 1

$$\Rightarrow \quad \mathbb{E}\|X\hat{\Theta} - X\Theta^*\|_F^2 \lesssim \sigma^2 k T \log d$$

We can do much better than this!

Using the prior information that Θ^* is low-rank, we can get

$$\mathbb{E}\|X\hat{\Theta} - X\Theta^*\|_F^2 \lesssim \sigma^2 \,\mathrm{rank}(\Theta^*) \max\{d, T\}$$

- Estimator for Problem 3:

$$\hat{\Theta}^{RK} := \underset{\Theta \in \mathbb{R}^{d\times T}}{\operatorname{argmin}} \left\{ \frac{1}{n}\|Y - X\Theta\|_F^2 + 2\lambda^2 \mathrm{rank}(\Theta) \right\}$$

Exercise 1: given Y, find explicit formula for

$$\underset{\Theta : \mathrm{rank}(\Theta)\le k}{\operatorname{argmin}} \|Y - \Theta\|_F$$

Exercise 0: given diagonal Y, what is

$$\underset{\Theta : \mathrm{rank}(\Theta)\le k}{\operatorname{argmin}} \|Y - \Theta\|_F = [Y_{ij} 1_{|Y_{ij}|\ge\lambda}]_{ij}$$

if U, V are orthogonal

$$\|UAV\|_F = \|A\|_F$$

Exercise 2: given Y and X, find explicit formula for

$$\operatorname{argmin}_{\Theta} \|Y - X\Theta\|_F$$

Θ : $\operatorname{rank}(\Theta) \le k$

$$\|Y - X\Theta\|_F^2 = \|Y - \bar{Y}\|_F^2 + \|\bar{Y} - X\Theta\|_F^2$$

$\bar{Y}$ ↑ projection of Y on $\operatorname{span}(X)$

Thm: Let

$$T := 4\sigma\sqrt{\frac{\log(12)\max\{d, T\}}{n}} + 2\sigma\sqrt{\frac{2\log\frac{1}{\delta}}{n}}$$

Then with probability $\ge 1 - \delta$,

$$\|X\hat{\Theta}^{RK} - X\Theta^*\|_F^2 \le 8n\operatorname{rank}(\Theta^*)T^2 \lesssim \sigma^2\operatorname{rank}(\Theta^*)\left(\max\{d, T\} + \log\frac{1}{\delta}\right)$$

↳ doesn't depend on X

Proof: optimality

$\Rightarrow$

$$\|Y - X\hat{\Theta}^{RK}\|_F^2 + 2nT^2\operatorname{rank}(\hat{\Theta}^{RK}) \le \|Y - X\Theta^*\|_F^2 + 2nT^2\operatorname{rank}(\Theta^*)$$

$\Leftrightarrow$

$$\|X\hat{\Theta}^{RK} - X\Theta^*\|_F^2 \le 2\langle E, X\hat{\Theta}^{RK} - X\Theta^*\rangle - 2nT^2\operatorname{rank}(\hat{\Theta}^{RK}) + 2nT^2\operatorname{rank}(\Theta^*) \quad (\#)$$

$$2\langle E, X\hat{\Theta}^{RK} - X\Theta^* \rangle \le 2\langle E, U \rangle^2 + \frac{1}{2}\|X\hat{\Theta}^{RK} - X\Theta^*\|_F^2 \quad (*)$$

where $$U := \frac{X\hat{\Theta}^{RK} - X\Theta^*}{\|X\hat{\Theta}^{RK} - X\Theta^*\|_F}$$

(Young's inequality: $2ab \le 2a^2 + \frac{1}{2}b^2$)

Assume for simplicity, X has full column-rank (result holds in general even if X is not full-rank; see refs in notes)

$$\langle E, U \rangle^2 \le \|E\|_{op}^2 \|U\|_{S_1}^2$$

$\|U\|_{S_1}^2 = \frac{\|U\|_{S_1}^2}{\|U\|_F^2}$ → C-S $\le \text{rank}(U)$

$$\le \|E\|_{op}^2 \, \text{rank}(U)$$

X is full-rank $\le \|E\|_{op}^2 \, \text{rank}(\hat{\Theta}^{RK} - \Theta^*)$

Subadditivity of rank → $\le \|E\|_{op}^2 \left(\text{rank}(\hat{\Theta}^{RK}) + \text{rank}(\Theta^*)\right)$

random matrix theory

$\Rightarrow \|E\|_{op}^2 \le n\tau^2$

p-Schatten norm:

$\|A\|_{sp} = \left(\sum_i \sigma_i^p\right)^{\frac{1}{p}}$ ← singular vals

$\|A\|_{s\infty} = \|A\|_{op}$ ← operator norm

$\|A\|_{s2} = \|A\|_F$

$\|A\|_{s1}$ = nuclear norm

matrix Hölder ineq: $p \in [1, \infty]$

$\langle A, B \rangle = \text{tr}(A^T B)$

$\le \|A\|_{sp} \|B\|_{sp'}$ ← Hölder conjugate

$\frac{1}{p} + \frac{1}{p'} = 1$

$$\langle E, U \rangle^2 \leq n\tau^2 \left(\text{rank}(\hat{\Theta}^{RK}) + \text{rank}(\Theta^*) \right)$$

But (*) (#)

$\Rightarrow$

$$\|X\hat{\Theta}^{RK} - X\Theta^*\|_F^2 \leq 4\langle E, U \rangle^2 - 4n\tau^2 \text{rank}(\hat{\Theta}^{RK}) + 4n\tau^2 \text{rank}(\Theta^*)$$

$$\leq 8n\tau^2 \, \text{rank}(\Theta^*)$$

■

Remark: L_1 counterpart also works; it replaces $\text{rank}(\Theta)$ in the objective by $\|\Theta\|_{S_1}$

- For Problem 1 and 2

$$\hat{\Theta} = \underset{\Theta : A(\Theta) = Y}{\text{argmin}} \; \|\Theta\|_{S_1}$$

↑ nuclear norm

Convex optimization